

INTRODUCTION TO INCIDENT RESPONSE:

- Security incident means anything that is suspicious like:
multiple failed logins
Malware detected
Data leak
Unauthorized access
Phishing attack
- Incident response is a process of handling a incident or security issue.
- When a security solution gives a alert, these alerts can be of two types:
 - 1 = False positive: A safer activity marked as harmful
 - 2 = True positive: A activity which is actually harmful and dangerous. These true positive alerts are also referred as incident.
- Incidents can be ~~was~~ classified on the basis of their severity level.

• Types of incidents:

- Phishing email:** In this type of incident a system is compromised by sending a phishing email with a malicious file attached to it.
- Malware infection:** Malware is a malicious program that can damage a system, network or application. In this a malware is injected by the attacker with the help of phishing email or anything else and this malware can be text, document, executables, etc.

- Data leaks can be unintentionally caused by human errors or misconfiguration whereas security breaches are done intentionally by an attacker.

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c) Security breaches: Security breaches arise when an unauthorized person gains access to an organization's confidential data.

d) Data leaks: Data leaks are incidents where the sensitive data of the organization or individual is exposed to unauthorized entities.

e) Insider attacks: Incidents or attacks, done by an employee or part of an organization. These attacks are done by insiders and can be harmful because insiders have more access to data.

(DoS attacks)

f) Denial of service attacks: In this incident, an attacker makes a website or server unavailable for real users by sending too many fake requests or we can say flood the system or server, so the server crashes or goes down.

• Incident response process:

There are two widely used incident response frameworks: SANS and NIST.

a) SANS incident response framework has 6 phases. Can be remembered by "PICKERL".

Preparation: This phase includes building the necessary resources to handle an incident such as having a plan, building a team, deploying security solutions.

Identification: This refers to looking for any abnormal activity that may indicate an incident.

Containment: In this, the impact of the attack is minimized by ~~compromising~~ isolating the victim's machine and disabling the compromised user account etc.

Eradication: In this phase the threat or malware is being removed from the attacked environment.

Recovery: In this phase the affected system perform recovery from backup or ~~the~~ rebuilding them so that they are ready to use.

Lessons Learned: In this a post incident review meeting is conducted to analyze the incident's root cause and improve the security to prevent future attacks.

6) NIST incident response framework has 4 phases:

Preparation → Detection and Analysis → Containment → Post incident
 , Eradication and Recovery Activity

• Incident response techniques:

It is harder to detect the abnormal behaviour and identify incidents manually. There are various security solutions that can detect incidents and even perform containment and eradication etc.

a) SIEM: Collects all logs in centralized location and correlates them to identify incidents.

b) AV (Antivirus): Antivirus detects known malicious files present on your system.

c) EDR: Endpoint Detection and response is deployed on every system. This can also contain and eradicate the threat.

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- What are playbooks? that must be followed
Playbooks are the step by step instructions, to handle a security incident.

Runbooks are the detailed step-by-step execution of specific steps during different incidents.